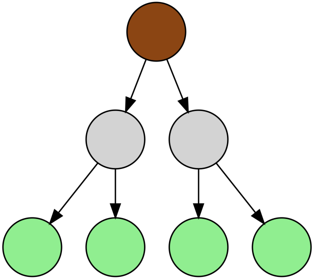
Decision Trees and Random Forest

Comp. Bio II

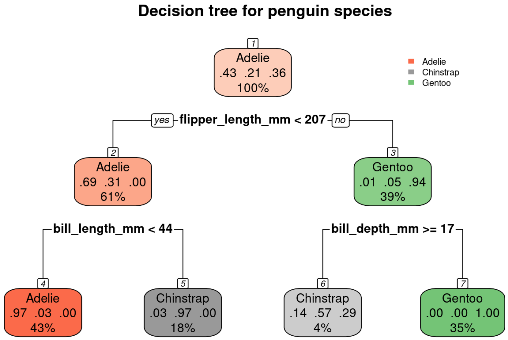
5th May, 2026

Trees



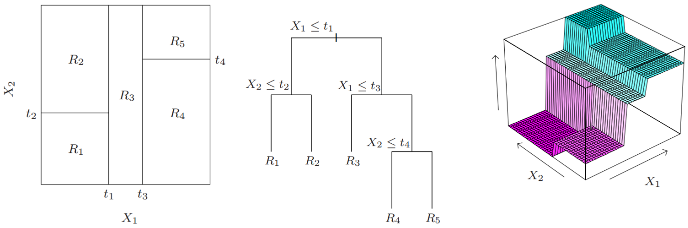
- A **tree** is a connected graph with no cycles.
- A **rooted tree** has a special node called the **root**.
- The root has no parent. All other nodes have exactly one parent.
- **Leaves** are nodes with no children.

Decision tree



- Internal nodes split the data based on feature values.
- Leaves assign a constant prediction (e.g. class label).

Decision tree - visualization



Hastie et al.: *The Elements of Statistical Learning*, 2009

- A decision tree models a piecewise constant function.
- Can be used for both regression and classification.

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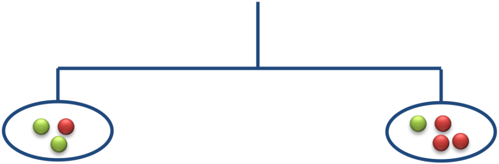
- Stop when a stopping criterion is reached (e.g. small nodes).
- Many variations exist (e.g. limiting depth, random feature selection).

Gini index



- Gini index is defined as

$$G = 1 - \sum_{k=1}^{K} p_k^2$$

where p_k is a fraction of k-th class (out of K classes) in the node.

- Equals the probability of misclassification (if labels are assigned according to class proportions).
- Used to train classification trees.

Bias-variance trade-off

- **Shallow trees**: low variance, high bias (may miss complex patterns).
- **Deep trees**: low bias, high variance (prone to overfitting).
- The trade-off can be controlled by model parameters (e.g. tree depth, minimum node size).

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- Can we get the best of both worlds?

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- **Bagging** (Bootstrap aggregating):
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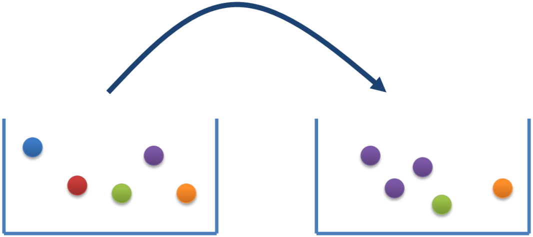
- **Boosting**:
  - Sequentially train *low variance, high bias* models (shallow trees).
  - Subsequent models learn to fix the mistakes of the previous ones.
  - Exploits dependence between models.

Bootstrapping



- Resampling from the original training set (with replacement).
    - Same number of observations.
    - Some observations may be sampled multiple times.
    - Some observations may not be sampled at all.
- Each resampling produces a slightly different version of the original data.

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- Additional randomness: at each tree split, consider only a random subset of features.
  - Reduces correlation between trees.
  - May also prevent trees from getting stuck in local optima during training.